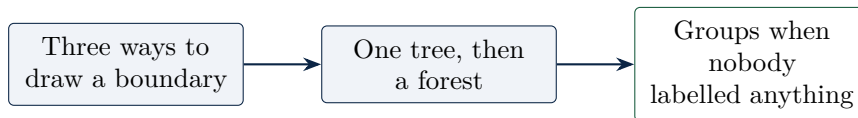


Machine Learning Operations

Lecture 04

A Toolbox of Models

Measure, do not guess



six models, one split, one table, and a written reason

Recap of Lecture 03 in three points. How much is regression; which one is classification. Always hide a test set, and score only on it. A score means nothing without a baseline to beat. We have one model and one honest number. We never asked whether a *different* model would do better. Today we ask, and we answer it with a measurement rather than an opinion.

Every score in this handout was measured on the course dataset, at the same split and seed as Lecture 03. Nothing is invented.

All sources verified 9 September 2026.

Contents

How this handout maps to the approved session plan	4
I Three Ways to Draw a Boundary	5
1 Classification is drawing a line	5
2 Logistic regression	5
3 k nearest neighbours	6
4 Support vector machines	6
5 Scale your inputs, or the biggest column decides everything	6
6 Common errors in choosing a classifier	7
II Trees and Forests	8
7 Rules a human can read	8
8 One tree is fragile; a forest is not	8
9 We measured, and the simplest model won	9
10 So which model does our project use?	9
11 Common errors with trees and forests	10
III Clustering Without Labels	11
12 When nobody labelled the data	11
13 k means, in four steps	11
14 Common errors in clustering	12
IV The Model Artifact, and AutoML	13
15 What a trained model actually is, once training ends	13
16 AutoML, in one page	13

V Consolidation	15
17 What our project gains today	15
18 Assignment questions, as released	15
19 Check your understanding	16
20 Vocabulary introduced today	16
VI The Week Ahead	18
21 Seven days, about seven hours	18
VII Resource Library	20
22 For this lecture specifically	20
23 Carried forward	20

List of Figures

1	The three classifiers of this lecture, and the single question that separates them. Everything else about them follows from how each decides where the boundary belongs.	5
2	Logistic regression outputs a probability, not a decision. The decision is the dashed line, and moving it is a business choice about which mistake you would rather make.	5
3	Nearest neighbours in one picture. Note the consequence for production: the whole training set must ship with the model, and every prediction is a search.	6
4	The widest gap, and the handful of points that determine it. Strong on small, clean datasets; slow on very large ones, because the search grows with the data.	6
5	The same six orders, before and after standardisation. Tree based models are almost unaffected by scaling; distance based ones such as nearest neighbours are transformed by it.	7
6	A decision tree over the delivery data. Readability is the selling point, and it is the thing you give up when you move to a forest.	8
7	Why averaging helps. A single tree's mistakes are systematic; fifty trees trained on different samples make different mistakes, and the average is steadier than any one of them.	8
8	The measured comparison. Our teaching data was generated from a straight line rule, so a straight line model should win, and it does. On real delivery data the relationships bend, and forests usually come out ahead.	9
9	The two halves of machine learning. The right hand column is not a harder version of the left; it answers a different kind of question and is judged differently.	11
10	k means in four steps. It stops when the centres stop moving. Note what the algorithm never does: it never tells you what k should be. That decision is yours, and it is not a technical one.	11
11	What has to travel with a model for it to be useful to anyone but you. Every item on the right hand list is a question somebody will ask in an incident review.	13
12	Still a notebook, and now a notebook that has compared its options honestly and recorded why it chose what it chose.	15

How this handout maps to the approved session plan

Topic in this handout	Session plan location	Note
Model families: logistic regression, nearest neighbours, support vector machines, trees, forests, k means	Session 5	The session plan lists model families under the first machine learning bridge; the released deck places them here, and this handout follows the released material
Model artifacts and baselines	Session 6	<i>Absent from the released deck.</i> Added as Part IV of this handout, since the next practical persists a model to disk and needs the vocabulary
AutoML, by demonstration	Session 6	Also absent from the deck. Covered briefly in Part IV, at the level the session plan asks for
Evaluation metrics	Session 5	Covered in Lecture 03, and used throughout today

Two inconsistencies worth raising at the faculty sync

Course outcome tagging. The assignment frames in the released deck tag these three question sets as CO1, while the accompanying assignments file tags them CO2. Applying the models to the running project is CO2 work, so this handout uses CO2. One of the two source files needs correcting, or attainment will not compute cleanly.

The project arc. The deck's arc figure shows Kubernetes as one of the eight boxes. In the approved plan Kubernetes is conceptual only and Docker Compose is the tool the course actually uses. The arc in this handout reflects the approved plan.

The taught hour, minute by minute

0 to 4	Recap in three points, then Section 1, classification as drawing a boundary
4 to 16	Sections 2 to 4, the three classifiers. One figure each, no mathematics
16 to 22	Section 5, scaling. Do not skip it; it explains two of the three assignments
22 to 30	Sections 7 and 8, the tree and then the forest
30 to 42	Section 9, the measured table. This is the centre of the lecture. Let them predict the winner before you reveal it
42 to 48	Section 10, which model the project carries forward, and the honest reason
48 to 58	Sections 12 to 14, unsupervised learning and k means
58 to 60	Section 17, what the project gains, and the summary
Part IV, the model artifact and AutoML, is the second slot or self study.	

PART I

Three Ways to Draw a Boundary

1 Classification is drawing a line

Classification is, at heart, drawing a boundary. On one side, this order will be late. On the other, this one is fine.

The three classifiers below differ in *one thing only*: how they decide where that boundary goes. You are not expected to master the mathematics today. You are expected to know what each one does, and when each is a sensible first choice.

LOGISTIC REGRESSION Fits a probability “83 percent chance this one is late” <i>you choose the cut off</i>	NEAREST NEIGHBOURS Asks the neighbours “what were the five most similar orders?” <i>stores the training set</i>	SUPPORT VECTOR MACHINE Finds the widest gap “which line separates them with most room?” <i>only edge points matter</i>
--	--	---

Same data, same question, three different boundaries. None of the three is the correct answer in general, which is precisely why Section 9 measures them rather than argues about them.

Figure 1: The three classifiers of this lecture, and the single question that separates them. Everything else about them follows from how each decides where the boundary belongs.

2 Logistic regression

Logistic regression

Despite the name, it *classifies*. It gives the probability that something belongs to a group, a number between 0 and 1.

An output of 0.83 means an eighty three percent chance that this order is late. You choose the cut off: above 0.5, call it late. It is fast, and you can read which inputs mattered, which is why it is the sensible first classifier to try, for the same reason a straight line is the sensible first regressor.

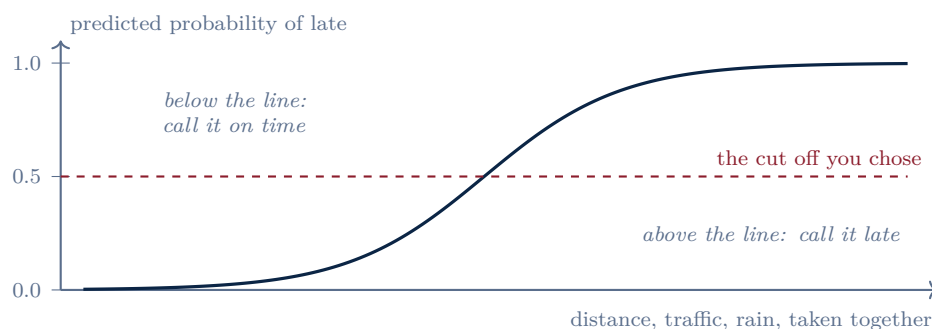


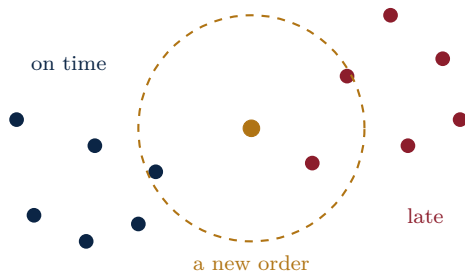
Figure 2: Logistic regression outputs a probability, not a decision. The decision is the dashed line, and moving it is a business choice about which mistake you would rather make.

The subtlety students miss

Because the cut off is yours, a single trained model gives you a whole family of classifiers. Lower it to 0.3 and you catch more late orders and raise more false alarms. Raise it to 0.7 and the reverse. Nothing about the model changed; only the threshold did.

3 k nearest neighbours

It does not learn a rule at all. It remembers every training row, and when a new order arrives it finds the k most similar past orders and takes their majority answer.



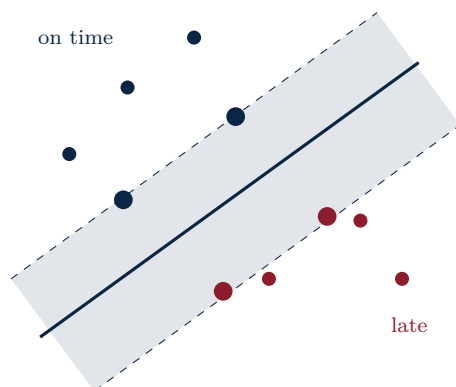
The circle holds the five nearest past orders. Four of them were late, so this one is called late. No rule was ever learned; the training data is the model.

Figure 3: Nearest neighbours in one picture. Note the consequence for production: the whole training set must ship with the model, and every prediction is a search.

4 Support vector machines

Support vector machine

It looks for the line with the widest empty gap on either side, on the argument that the boundary with the most room is the one most likely to hold on data it has not seen.



The larger points sit on the edge of the gap. They are the support vectors, and they alone define the boundary. Delete any other point and the line does not move.

A kernel lets the same idea bend, so the boundary need not be straight.

Figure 4: The widest gap, and the handful of points that determine it. Strong on small, clean datasets; slow on very large ones, because the search grows with the data.

5 Scale your inputs, or the biggest column decides everything

Two of the three classifiers above measure distance between rows. If one column runs from 0 to 12 and another from 0 to 1, the first column dominates every distance, and the model is effectively ignoring the second.

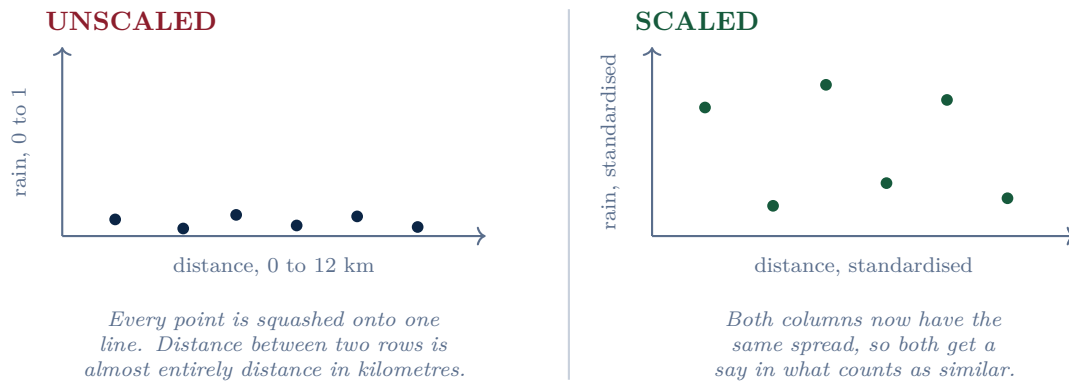


Figure 5: The same six orders, before and after standardisation. Tree based models are almost unaffected by scaling; distance based ones such as nearest neighbours are transformed by it.

The one line that fixes it, and where it must go

Standardise each column to a mean of zero and a standard deviation of one, using `StandardScaler`. Fit the scaler on the *training* rows only, then apply it to the test rows. Fitting it on everything is the leakage error from Lecture 03, and a Pipeline is what stops you making it.

6 Common errors in choosing a classifier

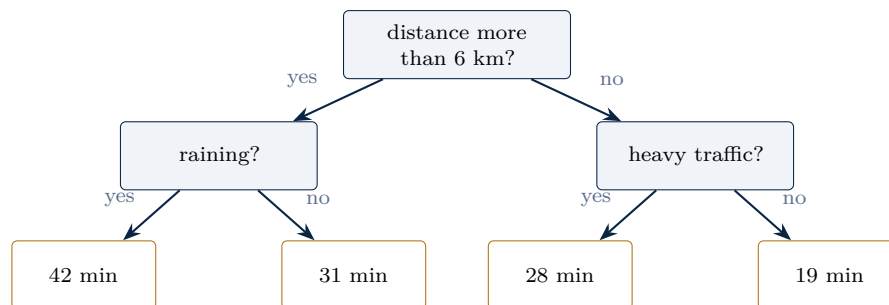
The error	Why it costs you
Reaching for the fanciest model first	Start simple. You need something to beat, and often the simple thing wins outright
Using nearest neighbours without scaling	Distance in kilometres and a rain flag are not comparable quantities
Reading a probability of 0.83 as certainty	It is a score, not a promise, and the cut off is yours to defend
Judging models on different splits	Then you compared the splits, not the models
Forgetting that nearest neighbours stores everything	Fast to train, slow to serve, and the whole dataset ships with the model

PART II

Trees and Forests

7 Rules a human can read

A decision tree asks yes or no questions until it reaches an answer. Its great virtue is that you can show it to a restaurant manager and they will follow it.



Every prediction is a path from the top to a leaf, and the path is the explanation. No other model in this lecture gives you that for free.

Figure 6: A decision tree over the delivery data. Readability is the selling point, and it is the thing you give up when you move to a forest.

8 One tree is fragile; a forest is not

Random forest

Train many different trees, each on a random part of the data, then take the average of their answers.

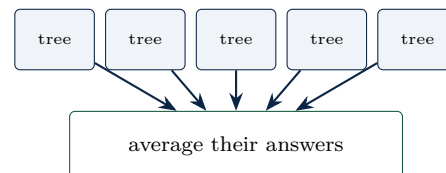
A single expert can be wrong. Ask fifty and average, and that is a forest. It is much harder to overfit than one deep tree, and it handles bends and interactions without being told about them. What you lose is the single readable rule.

ONE DEEP TREE

sees every row
memorises the quirks
0.00 on train, 3.43 on test

Measured in Lecture 03. Perfect on what it has seen, worst on what it has not.

A FOREST OF TREES



Each tree sees a random part. Their errors point in different directions and cancel.

Figure 7: Why averaging helps. A single tree's mistakes are systematic; fifty trees trained on different samples make different mistakes, and the average is steadier than any one of them.

9 We measured, and the simplest model won

Six models, one split, scored with mean absolute error in minutes on our 120 test orders.

On our 120 test orders	MAE, minutes
always guess the average	10.32
linear regression	1.92
random forest, 150 trees	2.43
k nearest neighbours, k equals 5	2.91
decision tree, depth 8	3.22
support vector regression	3.70

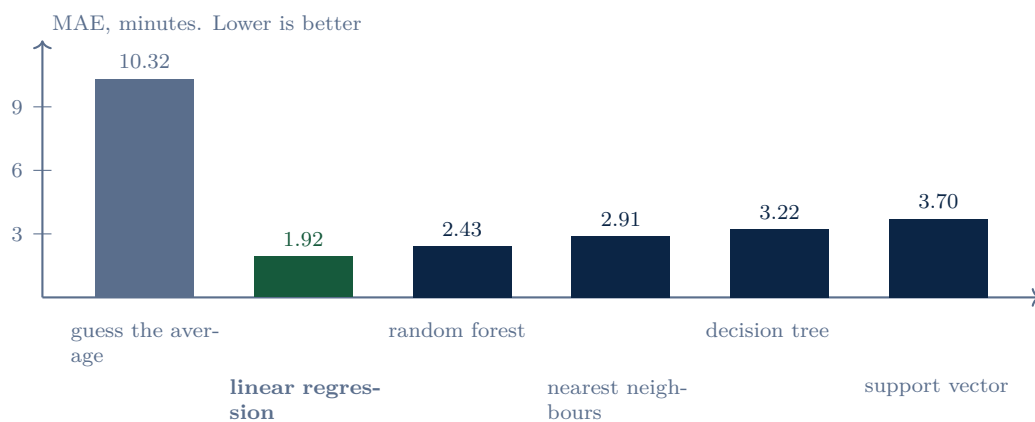


Figure 8: The measured comparison. Our teaching data was generated from a straight line rule, so a straight line model should win, and it does. On real delivery data the relationships bend, and forests usually come out ahead.

The lesson, which is not about linear regression

The fanciest model is not automatically the best. This is not a slogan; it is a theorem. Wolpert and Macready showed that any algorithm's elevated performance on one class of problems is paid for by worse performance on another, so no algorithm is best everywhere. Which one wins on *your* data is a question you answer by measuring, and the measurement takes about ninety seconds.

10 So which model does our project use?

On this dataset: linear regression, and we say so out loud rather than hiding it. From Lecture 11 onward the project carries a **random forest**.

Why the forest, then?

Because the rest of this course is about the machinery around a model: tracking, serving, shipping, watching. A forest exercises that machinery honestly. It has settings worth tuning, so experiment tracking has something to track. It produces a file worth versioning, so the registry has something to register. A linear model with four coefficients would make every later lecture look trivial for the wrong reason.

Saying this out loud is the point. Never quietly pick the model that flatters the story.

11 Common errors with trees and forests

The error	The correction
Letting a tree grow without a depth limit	It memorises the training rows. We measured exactly this in Lecture 03
Assuming more trees is always better	Past a point you pay in time and gain nothing measurable
Reporting a forest's score without a baseline	The baseline is what turns a number into evidence
Believing the fancier model must win	Measure it. Ours did not

PART III

Clustering Without Labels

12 When nobody labelled the data

Everything so far needed a right answer to learn from. Sometimes there is no label at all. Group our customers into similar types, says the brief. Into what types? Nobody knows yet, and that is the problem to be solved rather than the input to it.

SUPERVISED

You are given inputs *and* answers
 The model learns the mapping
 Success is measurable
 against the answers

Lectures 03 and 04 so far. Delivery minutes were always in the data.

UNSUPERVISED

You are given inputs only
 The model finds structure
 Success is a judgement, not a score

There is nothing to be accurate against, so accuracy cannot be quoted.

Figure 9: The two halves of machine learning. The right hand column is not a harder version of the left; it answers a different kind of question and is judged differently.

13 k means, in four steps

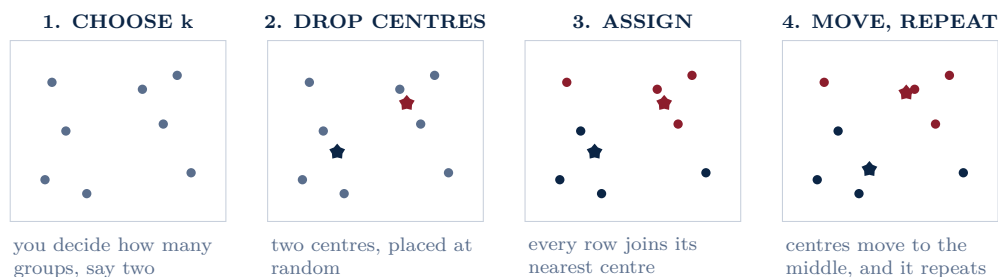


Figure 10: k means in four steps. It stops when the centres stop moving. Note what the algorithm never does: it never tells you what k should be. That decision is yours, and it is not a technical one.

Two things k means cannot do for you

It cannot choose k. Run it with k equal to 2, 3 and 4 and you get three different, equally valid answers. Which one you report depends on what the restaurant manager can act on, not on the mathematics.

It cannot tell you the groups are real. Clusters are a suggestion about your data, not a fact about your customers. Different algorithms on the same data find different groups, and none of them is wrong.

14 Common errors in clustering

The error	The correction
Expecting k means to tell you k	It cannot. That is your decision, and it should be defensible in business terms
Clustering unscaled data	The column with the biggest numbers silently decides everything, exactly as in Section 5
Treating clusters as truth	They are a suggestion, not a fact about your customers
Scoring clusters with accuracy	There are no labels to be accurate against

PART IV

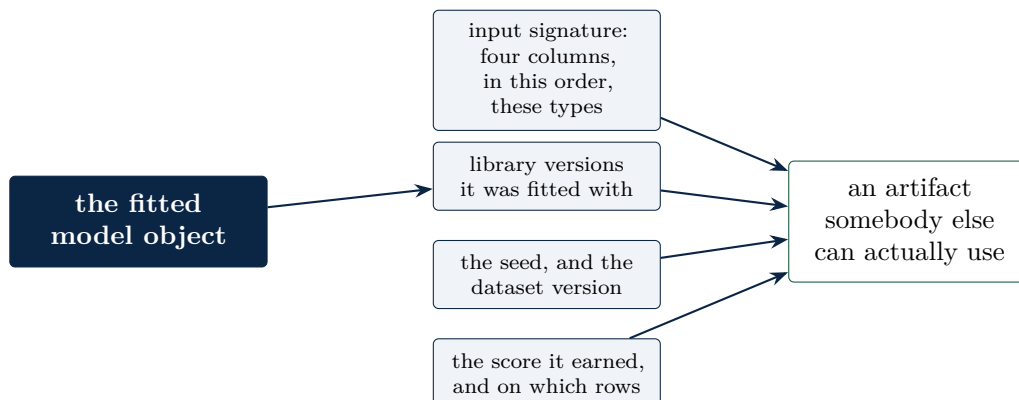
The Model Artifact, and AutoML

This part is not in the released slides. It covers the two remaining topics the approved session plan places in this session, and the next practical needs both.

15 What a trained model actually is, once training ends

A trained model is an object in memory. Close the notebook and it is gone. To use it tomorrow, or on another machine, or inside a service, it has to become a file. That file is the **model artifact**.

```
import joblib
joblib.dump(model, "model.joblib")    # write it
model = joblib.load("model.joblib")   # read it back
```



The weights alone are not enough. An artifact that arrives without its signature, its versions and its provenance is a file nobody can safely deploy, and this is exactly what the tracking server and the registry will store for you from Lecture 11.

Figure 11: What has to travel with a model for it to be useful to anyone but you. Every item on the right hand list is a question somebody will ask in an incident review.

A caution about serialised models

A serialised model is not a portable, permanent format. It records the state of a Python object, so loading it in a different library version may fail or, worse, may succeed and behave differently. Pin your versions, record them beside the artifact, and never load a model file from a source you do not trust: loading executes code.

16 AutoML, in one page

AutoML tools search over model families and settings automatically, train dozens of candidates, and hand you the best performer along with the code that produced it.

What it genuinely does	What it does not do
Searches families and settings faster than you can by hand	Choose the question, regression or classification, or define what a good answer is
Gives a strong baseline in minutes	Tell you whether the winning score is trustworthy, or whether the split leaked
Produces runnable code you can inspect	Take responsibility for the deployment, the drift, or the fairness of what it found

The honest framing for this course: AutoML competes with the ninety seconds you spent in Section 9, not with the twelve weeks you will spend on everything else. It automates the smallest box in the lifecycle diagram from Lecture 01.

PART V

Consolidation

17 What our project gains today



The project now has a measured comparison rather than a guess: six models, one split, one table, and a written reason for the model we carry forward. That table is the first thing the tracking server will remember for us in Lecture 11.

Figure 12: Still a notebook, and now a notebook that has compared its options honestly and recorded why it chose what it chose.

18 Assignment questions, as released

Set 1 • Three classifiers • CO2

Q1 (Understand). Explain in one sentence each how logistic regression, k nearest neighbours and a support vector machine decide where the boundary goes. Then say which of the three stores the whole training set.

Q2 (Apply). A nearest neighbours model works well on a test set where all inputs are in the range 0 to 1, but badly once you add a `distance_km` column ranging 0 to 12. Explain why, and state the one preprocessing step that fixes it.

Q3 (Build). Turn the delivery time problem into a classification problem: label an order late if `delivery_min` is above 40. Train logistic regression and nearest neighbours on the same split, report the accuracy of each, and write two lines on which you would choose and why.

Set 2 • Trees and forests • CO2

Q1 (Understand). Explain how a random forest differs from a single decision tree, and give one advantage and one disadvantage of the forest.

Q2 (Apply). Look at the measured table in this lecture. A teammate says the random forest is the best model here. State whether that is true for this dataset, quote the two numbers that settle it, and explain in two sentences why the result came out that way.

Q3 (Build). Train a decision tree on the delivery time data at depths 2, 4, 8 and unlimited. Record train and test MAE at each depth in a table, and write two lines saying which depth you would ship and how you can tell.

Set 3 • Clustering without labels • CO2

Q1 (Understand). Explain the difference between supervised and unsupervised learning, then describe the four steps of k means in your own words.

Q2 (Apply). A team clusters delivery records using `distance_km`, ranging 0 to 12, and `prep_time_min`, ranging 5 to 30, and finds the groups depend almost entirely on preparation time. Explain why, and state the fix.

Q3 (Build). Run k means on the delivery time data with k equal to 2, 3 and 4, using only

`distance_km` and `traffic_level`. For each k , describe in one line what the groups seem to represent, and say which k you would report to a restaurant manager and why.

19 Check your understanding

1. Name the single thing that separates logistic regression, nearest neighbours and a support vector machine.
2. Logistic regression outputs 0.83. What exactly does that number mean, and what still has to be decided before it becomes a prediction?
3. Which of today's models ships the entire training set to production, and why does that matter?
4. What are support vectors, and what happens to the boundary if you delete a point that is not one?
5. Give one model family that is almost unaffected by scaling, and one that is transformed by it.
6. Where must a scaler be fitted, and what is the name of the error you make if you fit it elsewhere?
7. State one advantage and one disadvantage of a forest over a single tree.
8. On our data the linear model beat the forest. Give the two numbers, and explain why the result came out that way.
9. Why does the project carry a random forest from Lecture 11, when linear regression scored better?
10. In k means, which decision does the algorithm never make for you?
11. Why can accuracy not be quoted for a clustering result?
12. Name three things that must travel with a model artifact besides the weights.

20 Vocabulary introduced today

Term	Meaning
Logistic regression	A classifier that outputs the probability of belonging to a group
Decision threshold	The cut off you choose to turn a probability into a decision
k nearest neighbours	A classifier that stores the training set and answers by majority of the k most similar rows
Support vector machine	A classifier that finds the boundary with the widest empty gap on either side
Support vectors	The points on the edge of that gap, which alone determine the boundary
Kernel	The trick that lets a support vector machine draw a curved boundary
Decision tree	A model of yes or no questions ending in an answer, readable by a human
Random forest	Many trees, each on a random part of the data, with their answers averaged
Standardisation	Rescaling a column to mean zero and standard deviation one
Supervised learning	Learning from inputs together with the right answers

Term	Meaning
Unsupervised learning	Finding structure in data that nobody has labelled
k means	An algorithm that groups unlabelled rows around k moving centres
Model artifact	The saved model file, plus the metadata needed to use and trace it
AutoML	Automated search over model families and settings

PART VI

The Week Ahead

21 Seven days, about seven hours

Day 1 • The families, from the source • 60 minutes

Read. The supervised learning chapter of the scikit-learn user guide: linear models, nearest neighbours, support vector machines and ensembles. https://scikit-learn.org/stable/supervised_learning.html

Produce. One line per family on what it assumes about the data. The assumption is the reason it wins or loses.

Day 2 • Why scaling changes the answer • 45 minutes

Read and run. The scikit-learn example, Importance of Feature Scaling. It shows that tree based models are almost unaffected by scaling, while nearest neighbours models fit a completely different model on unscaled data. https://scikit-learn.org/stable/auto_examples/preprocessing/plot_scaling_importance.html

Produce. Run nearest neighbours on our delivery data with and without a scaler, and record both scores.

Day 3 • Reproduce the table • 60 minutes

Do. Rebuild Section 9 yourself: six models, one split, seed 42, MAE on the test set. Do not copy the numbers from this handout.

Produce. Your own version of the table. If any number differs from ours, find out why before you go further; that is the whole skill.

Day 4 • Depth and overfitting • 45 minutes

Do. Assignment Set 2, Q3: a decision tree at depths 2, 4, 8 and unlimited, with train and test MAE at each depth.

Produce. The table, plus one sentence identifying where the two columns start to disagree. That point is the complexity curve from Lecture 03, measured on your own data.

Day 5 • Clustering, and its limits • 60 minutes

Read. The scikit-learn example, Comparing different clustering algorithms on toy datasets. Same data, several algorithms, several different answers, and none of them wrong. https://scikit-learn.org/stable/auto_examples/cluster/plot_cluster_comparison.html

Produce. Two sentences on what it would mean to say a clustering is correct.

Day 6 • No free lunch • 45 minutes

Read. Wolpert and Macready, *No Free Lunch Theorems for Optimization*, IEEE Transactions on Evolutionary Computation, 1(1), 67 to 82, 1997. Read the abstract and the introduction only.

Any elevated performance over one class of problems is offset by performance over another.

Produce. One paragraph connecting the theorem to why your table in Day 3 could not have been predicted in advance.

Day 7 • Save the model properly • 60 minutes

Do. Persist your best model with joblib, then load it in a *fresh* session and reproduce one prediction exactly.

Produce. A short note recording the four things from Part IV that travel with the artifact: input signature, library versions, seed and dataset version, and the score with the rows it was measured on. That note becomes your first model card.

If you only have one hour all week

Day 3, then Day 7. Reproducing the table is what makes the lesson yours rather than ours, and persisting the model is what the next practical assumes you can already do.

PART VII

Resource Library

All links verified on 9 September 2026.

22 For this lecture specifically

Resource	Where, and why
scikit-learn, supervised learning user guide	https://scikit-learn.org/stable/supervised_learning.html Every family in this lecture, from the source
scikit-learn, Importance of Feature Scaling	https://scikit-learn.org/stable/auto_examples/preprocessing/plot_scaling_importance.html Why Section 5 matters, demonstrated rather than asserted
scikit-learn, comparing clustering algorithms	https://scikit-learn.org/stable/auto_examples/cluster/plot_cluster_comparison.html Same data, several algorithms, several answers
scikit-learn, choosing the right estimator	https://scikit-learn.org/stable/machine_learning_map.html The map from Lecture 03, now with names you recognise
Wolpert and Macready, No Free Lunch Theorems for Optimization, 1997	IEEE Transactions on Evolutionary Computation 1(1), 67 to 82. https://doi.org/10.1109/4235.585893 The formal reason to measure rather than assume
Google, Rules of Machine Learning	https://developers.google.com/machine-learning/guides/rules-of-ml Rule 4: keep the first model simple and get the infrastructure right

23 Carried forward

Resource	Where
scikit-learn, common pitfalls and recommended practices	https://scikit-learn.org/stable/common_pitfalls.html
Kapoor and Narayanan, leakage and the reproducibility crisis	https://doi.org/10.1016/j.patter.2023.100804

Resource	Where
Sculley et al., Hidden Technical Debt in Machine Learning Systems	https://proceedings.neurips.cc/paper_files/paper/2015/file/86df7dcfd896fcdf2674f757a2463eba-Paper.pdf
Sato, Wider and Windheuser, Continuous Delivery for Machine Learning	https://martinfowler.com/articles/cd4ml.html
Amershi et al., Software Engineering for Machine Learning, ICSE 2019	https://www.microsoft.com/en-us/research/wp-content/uploads/2019/03/amershi-icse-2019_Software_Engineering_for_Machine_Learning.pdf
Machine Learning in Production, DeepLearning.AI, free to audit	https://www.coursera.org/learn/introduction-to-machine-learning-in-production
MLOps Zoomcamp, DataTalks.Club	https://github.com/DataTalksClub/mlops-zoomcamp
Made With ML	https://madewithml.com
Stanford CS 329S	https://stanford-cs329s.github.io

Our simplest model won, and we published that.

Next lecture. Lecture 05, version control philosophy. Today's winning model came from code, and if the code is not saved properly the winner cannot be rebuilt. Git for code, collaborative workflows with branches and pull requests, and what belongs in a repository and what does not. Bring the repository you cloned in the first practical.